

Exercise 3.02

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

part a

```
In [2]: theta = np.linspace(0,2*np.pi)

x = 2*np.cos(theta)+np.cos(2*theta)
y = 2*np.sin(theta) - np.sin(2*theta)
```

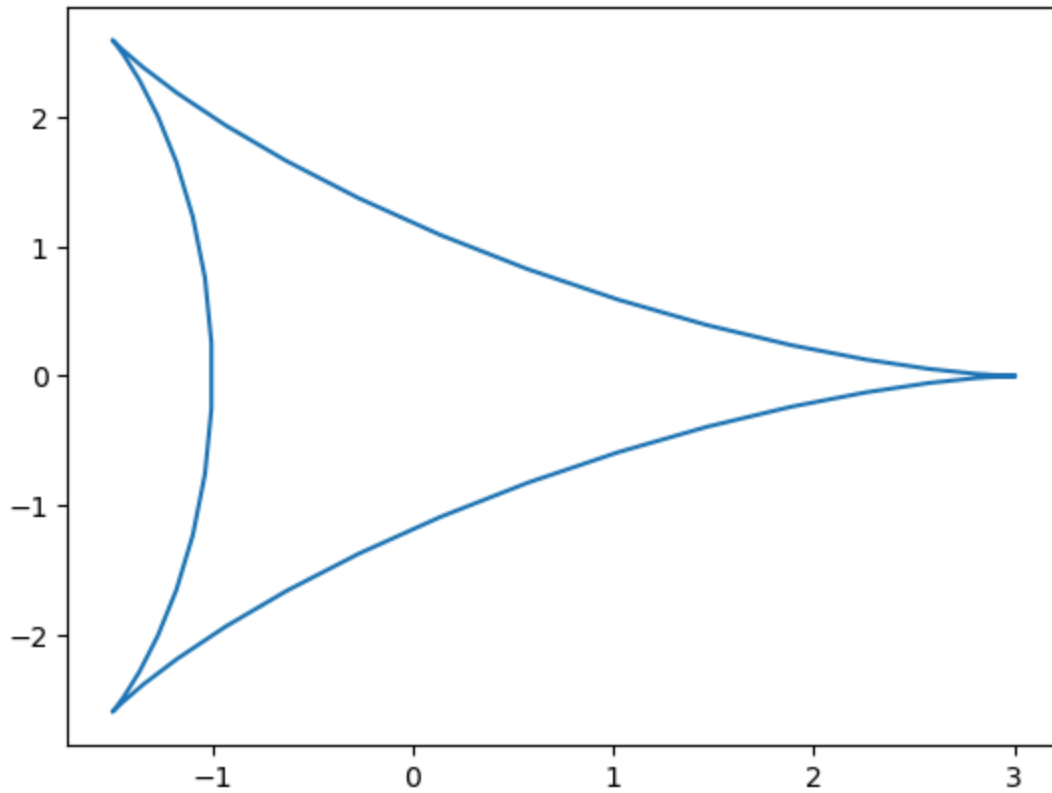
```
In [5]: y
```

```
Out[5]: array([ 0.00000000e+00,  2.09973946e-03,  1.65916158e-02,  5.48514592e-02,
                1.26292341e-01,  2.37553208e-01,  3.91878885e-01,  5.88735053e-01,
                8.23686220e-01,  1.08854725e+00,  1.37180080e+00,  1.65925535e+00,
                1.93490221e+00,  2.18191685e+00,  2.38373956e+00,  2.52516524e+00,
                2.59337087e+00,  2.57881293e+00,  2.47593511e+00,  2.28363857e+00,
                2.00548243e+00,  1.64959896e+00,  1.22832697e+00,  7.57584262e-01,
                2.56017602e-01, -2.56017602e-01, -7.57584262e-01, -1.22832697e+00,
               -1.64959896e+00, -2.00548243e+00, -2.28363857e+00, -2.47593511e+00,
               -2.57881293e+00, -2.59337087e+00, -2.52516524e+00, -2.38373956e+00,
               -2.18191685e+00, -1.93490221e+00, -1.65925535e+00, -1.37180080e+00,
               -1.08854725e+00, -8.23686220e-01, -5.88735053e-01, -3.91878885e-01,
               -2.37553208e-01, -1.26292341e-01, -5.48514592e-02, -1.65916158e-02,
               -2.09973946e-03,  0.00000000e+00])
```

```
In [9]: fig0 = plt.figure()
ax0 = fig0.add_subplot()

ax0.plot(x, y)
```

```
Out[9]: [<matplotlib.lines.Line2D at 0x7dd449737a70>]
```



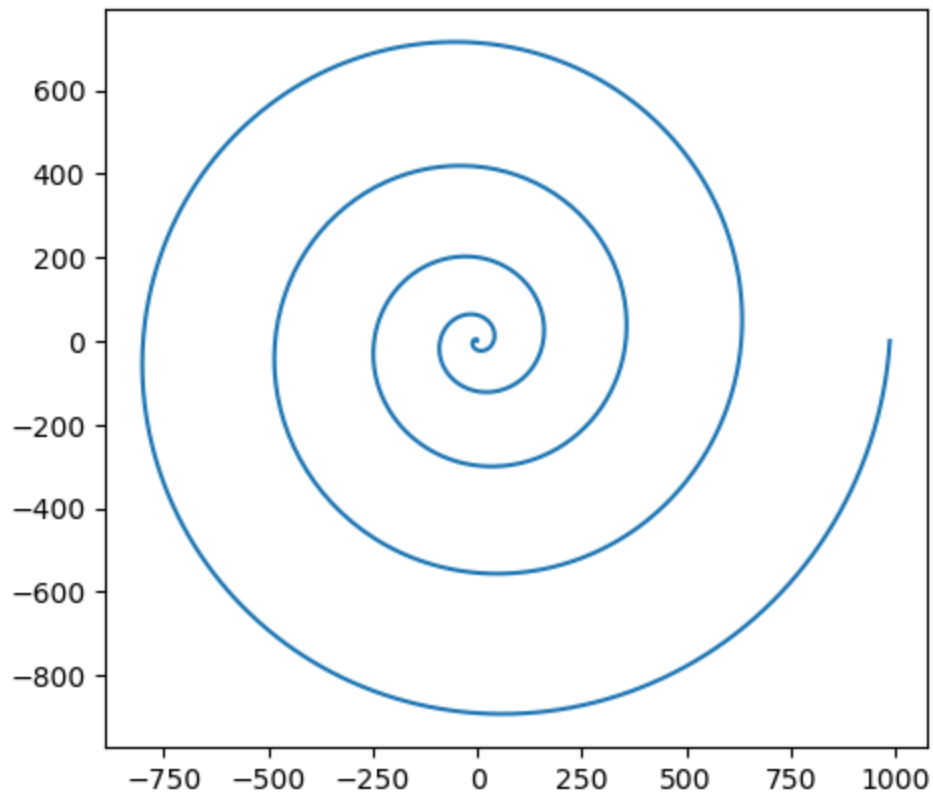
part b

```
In [40]: theta = np.linspace(0, 10*np.pi, 1000)
r = theta**2
x = r*np.cos(theta)
y = r*np.sin(theta)
```

```
In [41]: fig1 = plt.figure()
ax1 = fig1.add_subplot(aspect="equal")

ax1.plot(x,y)
```

```
Out[41]: [<matplotlib.lines.Line2D at 0x7dd43fb384d0>]
```



part c

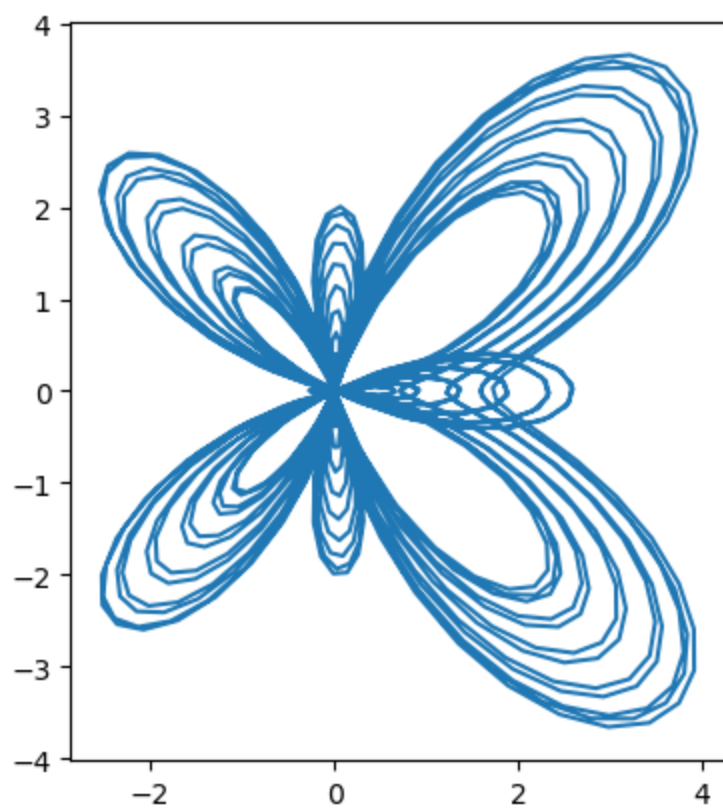
```
In [42]: theta = np.linspace(0, 24*np.pi, 1000)

r = np.exp(np.cos(theta)) - 2*np.cos(4*theta) + np.sin(theta/12)

x = r*np.cos(theta)
y = r*np.sin(theta)

In [43]: fig2, ax2 = plt.subplots(subplot_kw = {'aspect': 'equal'})

ax2.plot(x, y)
plt.show()
```



In []: